Recommendation for Modelling Transport Mode Choice in PRISM-games

Short technical report

1. Purpose of the model

The intended model represents a transport system in which a manager chooses transport policies, users choose travel modes, and the environment affects the system through weather, road closures, traffic incidents, infrastructure failures, and congestion.

The recommended modelling formalism is a turn-based stochastic multi-player game, implemented in PRISM-games as an smg, rather than a concurrent stochastic game. This choice reflects the information structure of the transport scenario rather than a limitation of the tool.

2. Recommended phase structure

The recommended structure is:

```
Manager -> Observable environment -> Users -> In-journey environment -> System update
```

More explicitly:

```
1. Manager sets policy
2. Observable environmental conditions are realised
3. Users choose travel mode
4. In-journey disturbances occur
5. Costs, delays, congestion, emissions and other rewards are updated
```

This structure gives a clear information pattern: users observe policy and visible conditions, but they do not observe future incidents. For example, users may reasonably observe rain before deciding whether to cycle, but they should not know in advance whether a traffic accident or sudden road closure will occur during their journey.

3. Motivation for using a turn-based model

A turn-based model is preferred because the transport scenario has a natural sequential interpretation. Timetables, fares, service frequency and pricing policies are usually set before travel decisions are made. Users then choose a mode given the available information. Subsequently, the journey unfolds under uncertainty.

This differs from a concurrent stochastic game, where the manager, users and environment would choose actions simultaneously. A CSG is useful when agents genuinely act concurrently, for example in real-time adaptive pricing, adaptive signalling, or demand-responsive transport. For the present model, however, simultaneity would obscure the information available to each decision-maker.

The main reason for avoiding a CSG here is interpretability. A turn-based model allows the order of information to be represented explicitly. The order of phases determines what each player can condition its strategy on.

4. Observable versus unobservable environmental events

The environment should be split into two components.

Observable pre-trip environment. These are conditions known before users choose their mode: weather, known roadworks, announced closures, published disruptions, service frequency, fare level and road-pricing level. These should occur before the user choice. This is suitable for a bike-versus-car decision, because current weather can strongly influence whether cycling is attractive.

In-journey environment. These are events that occur after users have committed to a mode: unexpected road closure, traffic accident, bus delay, bus overcrowding, signal failure, temporary cycle-lane obstruction and congestion propagation. These should occur after the user choice.

Sampling in-journey events before the user decision would leak information into the model. Since strategies can depend on the execution history, a user strategy could condition on a future incident if it had already been stored in the state. Thus, a sudden road closure should normally be modelled as:

```
Users choose mode -> Closure/incident occurs -> Delay and congestion update
```

whereas a planned closure should be modelled as:

```
Closure announced -> Users choose mode
```

5. Probabilistic or adversarial environment

The environment should initially be modelled probabilistically, not as a strategic adversary. This is appropriate if weather, traffic incidents and minor disruptions represent ordinary uncertainty rather than an intelligent opponent.

A strategic environment should be introduced only if the research question is explicitly about robustness, for example: can the manager guarantee acceptable service levels under worst-case disruptions? The trade-off is that an adversarial environment may be too pessimistic, since it can select severe weather, incidents, or closures whenever doing so frustrates the manager objective.

A useful compromise is a bounded adversarial environment, where severe events are allowed but constrained, for example: at most one major closure per day, severe weather cannot persist for more than k steps, or incident probability increases with congestion.

6. Recommended PRISM-games abstraction

A suitable high-level smg model would have the following strategic players:

```
player manager
  [set_normal], [set_high_freq], [set_low_fare], [set_road_charge]
endplayer

player users
  [choose_bike], [choose_car], [choose_bus]
endplayer
```

The environment can initially be represented by probabilistic commands rather than by a player. The phase structure would be:

```
global phase : [0..4] init 0;
// 0 = manager
// 1 = observable environment
// 2 = users
// 3 = in-journey environment
// 4 = update
```

The intended interpretation is: manager sets policy; weather or known disruption is sampled; users choose bike, car or bus; a journey disturbance is sampled; congestion, delay, emissions and rewards are updated. This gives a defensible separation between known conditions and future uncertainty.

7. Rewards and objectives

The model should include several reward structures rather than a single objective. At minimum: user travel time, user monetary cost, operator cost, manager revenue, emissions, congestion and service-level violations.

Typical questions include: can the manager keep expected congestion below a threshold; can users minimise expected travel cost; what is the expected emissions impact of subsidising cycling; how robust is a pricing policy against unexpected closures; and what is the probability of severe delay under a given strategy?

8. Final recommendation

The recommended model is a turn-based stochastic multi-player game with probabilistic environmental uncertainty, using the phase order:

```
Manager -> Observable environment -> Users -> In-journey environment -> Update
```

This choice is motivated by three considerations. First, it matches the natural information structure of transport decisions: policies and visible weather are known before travel, but incidents during the journey are not. Second, it avoids information leakage that would arise if future disturbances were sampled before the user decision. Third, it gives a simpler and more interpretable model than a native CSG, while still allowing the manager and users to be treated as strategic players.

A CSG should only be used if the model is meant to represent genuinely simultaneous adaptation. An adversarial environment should only be used for explicit robustness analysis. For the first version of the model, the cleanest encoding is therefore: strategic manager, strategic users, probabilistic observed weather, probabilistic in-journey disturbances, and explicit rewards for cost, time, congestion and emissions.

References

Kwiatkowska, M., Norman, G., Parker, D., and Santos, G. PRISM-games 3.0: Stochastic game verification with concurrency, equilibria and time. CAV 2020.

Chen, T., Forejt, V., Kwiatkowska, M., Parker, D., and Simaitis, A. Automatic verification of competitive stochastic systems. Formal Methods in System Design, 2013.

Puterman, M. L. Markov Decision Processes: Discrete Stochastic Dynamic Programming. Wiley, 1994.